

Hoang Mai Duc Kien

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Education

VNUK – Institute for Research and Executive Education November 2023 – Expected June 2027

Data Science

GPA: 3.87/4.00

Relevant coursework: Probabilities & Statistics, Machine Learning & Deep Learning, Artificial Intelligence, Linear Algebra and Optimization for Data Science, Data Analytics for Life Science, Applied Forecasting

- GPA: 3.87/4.00
- 50% of the Merit Scholarship for Freshman year
- 100% of the Merit Scholarship for Sophomore year
- Best Intern for the Freshman Internship program

Phan Chau Trinh High School

November 2020 – May 2023

- GPA 8.9/10.0
- Third Prize, English Subject, City Excellent Student Contest
- Consolation Prize, Information Technology, School-level Excellent Student Contest

Work History

Hekate, JSC. – Da Nang, Viet Nam

October 2024 - December 2024

Creative Associate of Growth - Intern

- Gained in-depth knowledge of the AI industry and its operational frameworks through hands-on experiences.
- Contributed to creative growth initiatives, enhancing overall understanding of the industry.
- Recognized as one of the top 3 interns among 10 in the internship program.

Whitex Garment (Vietnam) Co. – Danang, Vietnam

January 2026 – April 2026

Data Analyst Intern

- Cleaned, transformed, and analyzed production, order, and energy-consumption datasets using Python, Pandas, and Excel to support operational decision-making.
- Conducted exploratory data analysis and Holt-Winters forecasting to identify production and energy-consumption trends.
- Analyzed production and energy metrics across 2023–2025, identifying a substantial increase in production volume and a decrease in energy consumption per piece.
- Proposed a centralized data repository and analytics dashboard to replace fragmented spreadsheets and improve reporting efficiency.

Projects & Achievements

DeepFake Detection System

November 2025 - Now

Python, PyTorch, Streamlit, Albumentations, OpenCV

Project Link: <https://github.com/ndrhmdk/deepfake-detection>

- Engineered a comprehensive deep learning pipeline to detect synthetic facial manipulations by implementing and fine-tuning CNN architectures including EfficientNet, Xception and ResNet18.
- Developed a real-time deployment interface using Streamlit, featuring an end-to-end inference engine that supports both image and video uploads with automated frame extraction.

- Implemented MTCNN for automated face cropping and Albumentations for complex data augmentation to improve model generalization.
- Integrated Explainable AI (xAI) by developing Grad-CAM visualization module to highlight mesoscopic facial artifacts, providing transparency into the model's decision-making process.
- Optimized data ingestion performance by architecting custom PyTorch DataLoaders that handle multi-modal inputs.
- Structured the project using a modular MVC-inspired approach, separating core logic into dedicated modules to ensure maintainability and scalability

Chest X-ray Classification System

November 2025 – January 2026

Python, PyTorch, Streamlit, Albumentations, OpenCV

Project Link: <https://github.com/ndrhmdk/deepfake-detection>

- Developed a high-performance deep learning pipeline to classify chest X-ray images into three diagnostic categories: Normal, COVID-19, and Pneumonia.
- Architected and compared multiple model variants, including a custom BaselineCNN and fine-tuned EfficientNet (B0) architectures, achieving optimised accuracy through transfer learning and hyperparameter tuning.
- Integrated Explainable AI (xAI) using Grad-CAM to generate heatmaps that visualize the specific regions of the lungs driving the model's prediction, ensuring clinical transparency and interpretability.

MOIRAI: Zero/Few-Shot Time Series Forecasting

November 2025 – January 2026

Python, PyTorch, Streamlit, Albumentations, OpenCV

Project Link: <https://github.com/ndrhmdk/foundation-moirai-time-series-forecasting>

- Evaluated Salesforce's Moirai in Zero-Shot and Few-Shot scenarios, benchmarking model variants (14M vs. 311M parameters) against traditional statistical baselines (ETS, Seasonal Naïve).
- Engineered a robust backtesting pipeline using rolling-origin cross-validation to assess model performance across financial and environmental domains without data leakage.
- Analyzed probabilistic forecasting performance by implementing advanced evaluation metrics including Pinball Loss, Interval Coverage, and Mean Absolute Scaled Error (MASE).
- Conducted rigorous statistical validation using the Wilcoxon Signed-Rank Test to determine the significance of improvements between foundation models and statistical benchmarks.
- Optimized experimental reproducibility by architecting a structured codebase with uv for dependency management and YAML-based configurations for flexible hyperparameter tuning.
- Evaluated Moirai against ETS and Seasonal Naïve baselines using rolling-origin backtesting on SPY and AQI datasets, with MAE, MASE, and probabilistic forecasting metrics.

Skills

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- **Programming Languages:** Python, C++, Java, JavaScript, HTML/CSS
 - **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-Learn, Data Visualization
 - **Data Wrangling:** Pandas, NumPy
 - **Critical Thinking**

Certifications

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| ● IELTS 7.5 | December 2025 |
| ● WorldQuant University - Applied Data Science Lab Certificate | May 2025 |